



PROJECT ATLAS

COMPLETE ASSEMBLY & OPERATION MANUAL

Build the rover from separated parts, wire it safely, power it on, validate it and operate it

AUDIENCE

Written for a careful first-time builder. No ROS or electronics experience is assumed.

IMPORTANT

The verified ATLAS installation is the authority. Match every printed model label and connector before applying power. Never infer a power pin from wire colour.

Prepared for Dhruv | Jetson Orin Nano Super 8GB | ROS 2 Humble

How to use this manual

Follow the numbered build sequence. Do not skip the inspection gates. A green PASS means continue; a red STOP means remove power and correct the issue.

Phase	What you do	Power state
A	Identify parts, label cables and prepare tools	Battery disconnected
B	Build chassis, wheels, steering and sensor mounts	Battery disconnected
C	Wire motors, data devices, I2C and power distribution	Battery disconnected
D	First power-up and software verification	Wheels lifted
E	Ground test, mapping and normal use	Clear area; E-stop accessible

Safety rules that cannot be bypassed

Rule	Reason
Disconnect the battery before changing wiring	Prevents shorts, motor movement and damaged boards
Fuse the battery output and keep the emergency stop in series with motor power	Stops high-current faults and provides physical control
Use a common ground but separate regulated logic and motor rails	Prevents unstable data and protects the Jetson
Never power servos or motors from Jetson GPIO	GPIO cannot supply motor current
Lift all four wheels for the first motor test	Incorrect polarity cannot drive the rover away
Keep LiDAR unobstructed and camera ribbon strain-free	Avoids false maps and CSI damage
Do not start autonomy when START_POSE BLOCKED	The planner believes the rover overlaps an obstacle

HIGH CURRENT

ATLAS battery and motor wiring can overheat or start a fire if polarity, fuse or conductor size is wrong. If the battery/BMS label differs from this rover's verified 12.6 V maximum, stop and obtain a qualified electrical review.

Tools, consumables and labels

Category	Required items
Hand tools	Metric hex keys, screwdrivers, 7/8/10 mm spanners, side cutters, pliers
Electrical	Digital multimeter, insulated ferrules, crimper, heat-shrink, cable ties, fuse holder
Mounting	M3/M4 hardware, nylon standoffs, thread-lock for metal-to-metal screws, foam tape
Data	USB cables, CSI ribbon, labelled JST/PH leads, USB-to-TTL only for L76K GNSS
Safety	Eye protection, nonconductive work surface, wheel blocks, accessible E-stop

Mandatory cable labels

Label	Connection
M1 FR	Motor 1 front-right
M2 FL	Motor 2 front-left
M3 BR	Motor 3 back-right
M4 BL	Motor 4 back-left
LIDAR USB	RPLIDAR A1 CP2102 adapter
YAHBOOM USB	Motor board CH340/CH341
RADAR TX/RX	Jetson UART pins 10/8
I2C 3/5	Jetson SDA/SCL shared sensor bus
GNSS USB	L76K through USB-to-TTL
CSI	IMX708 camera ribbon

LABEL FIRST

Apply labels at both ends before routing a cable. The Linux `/dev/ttyUSB` number can change; ATLAS uses stable `by-id` or `udev` names where available.

Major parts - identify before assembly



Jetson Orin Nano Super 8GB deployment kit

Main computer. Runs Ubuntu 22.04, ROS 2 Humble, Nav2, SLAM, AI and dashboards.

Source photo: ThinkRobotics product page



AM Robotics ROVER 4WDXL60R chassis

50 x 36 cm four-wheel-steering mechanical platform. Verify 36.7 cm front-to-rear wheel-centre spacing.

Source photo: AM Robotics product page

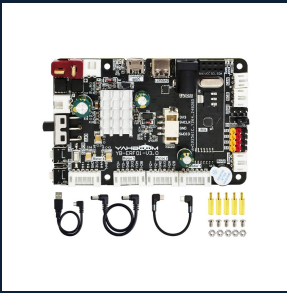


125 mm heavy-duty rubber wheels

Four 6 mm-shaft wheels. Installed circumference used by odometry: 0.392699 m.

Source photo: AM Robotics product page

Major parts - drive and navigation



Yahboom ROS robot control board

Four encoder motor channels, steering support and USB serial link to Jetson.

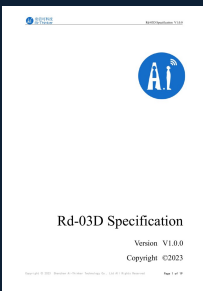
Source photo: Yahboom product page



SLAMTEC RPLIDAR A1

Primary 360-degree mapping and obstacle sensor. Mount level with a clear view.

Source photo: SLAMTEC product; photo via Robotopian listing

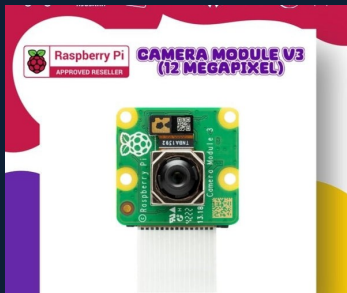


Ai-Thinker RD-03D 24 GHz radar

Motion-awareness layer. Connects by UART; does not replace LiDAR for mapping.

Source photo: Ai-Thinker official datasheet

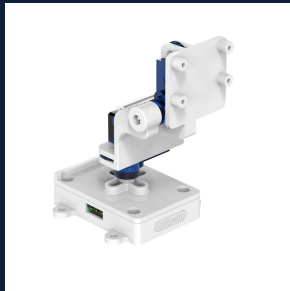
Major parts - vision and positioning



Raspberry Pi Camera Module 3 (IMX708)

12 MP CSI camera with powered autofocus and HDR capability; used for AI view and tracking.

Source photo: Raspberry Pi product; reseller product photo



Arducam B0283 upgraded pan/tilt

Two digital servos and I2C control board for camera direction. Separate regulated servo power.

Source photo: REES52 product page



Waveshare L76K multi-GNSS module

GPS, BeiDou, GLONASS and QZSS positioning. ATLAS connects it through USB-to-TTL, not occupied UART pins.

Source photo: Waveshare product; Pi Hut product photo

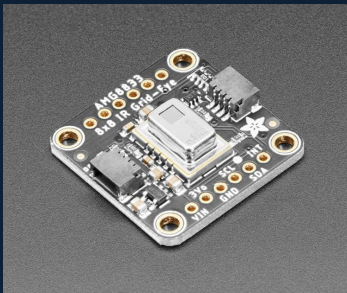
Major parts - environment and interface



BME680 environmental sensor

Outside temperature, humidity, pressure and gas resistance / air-quality trend over I2C.

Source photo: Adafruit product page



AMG8833 Grid-EYE 8x8 thermal sensor

Internal 64-pixel infrared temperature array over I2C.

Source photo: Adafruit learning guide



ESP32-S3 AI smart speaker board

Dual microphones, speaker, RGB ring and buttons for ATLAS voice interaction.

Source photo: ThinkRobotics product page

Major parts - operator interface and power



Elecrow CrowPanel Advanced 10.1 ESP32-P4

Removable wireless touch dashboard/controller. Jetson remains the safety and motor authority.

Source photo: Elecrow product page



DALY smart BMS and battery telemetry

Protects and reports the traction battery. Exact cell count/current rating must match the battery label.

Source photo: DALY product page

VERIFY MODEL

Internet photos are identification aids only. Product revisions can change connector position or pin order. Read the silkscreen and the supplied datasheet before wiring.

Mechanical assembly

Step 1 - Build the bare chassis

Assemble the ROVER 4WDXL60R lower frame on a flat table. Install cross-members loosely, square the frame, then tighten in a diagonal pattern. Keep the future battery area low and central.

Checkpoint	Pass condition
Frame	No twist; all four suspension/motor points sit evenly
Fasteners	Correct length; thread-lock only on metal-to-metal screws



Step 2 - Install motors and 125 mm wheels

Install M1 front-right, M2 front-left, M3 back-right and M4 back-left. Route encoder cables inward, away from tyres and steering links. Tighten each 6 mm wheel flange securely.

Checkpoint	Pass condition
Wheel order	M1 FR, M2 FL, M3 BR, M4 BL labels visible
Rotation	Every wheel turns freely by hand with power disconnected
Clearance	No cable, chassis edge or linkage rubs a wheel



Mechanical assembly - steering, compute and sensors

Step 3 - Install front and rear steering

Connect the front steering mechanism to the front axle and the rear mechanism to the rear axle. Set both mechanically near centre before attaching horns. Do not force a powered servo by hand.

Checkpoint	Pass condition
Centre	Both axles visually straight at neutral
Travel	Left/right motion does not bind or hit chassis
Direction	Front and rear steering respond in the configured four-wheel-steering sense

Step 4 - Mount Jetson and Yahboom controller

Mount the Jetson on insulated standoffs with fan airflow unobstructed. Mount the Yahboom controller where motor leads are short but high-current terminals cannot touch the chassis. Leave access to USB and the E-stop circuit.

Checkpoint	Pass condition
Jetson	No conductive contact below carrier board; fan clear
Controller	Terminals covered; USB strain relieved
Separation	Motor wires separated from CSI, I2C and antenna cables



Step 5 - Mount LiDAR

Place the LiDAR level and unobstructed. The verified centre is 30 cm behind the front chassis edge, which is 5 cm behind chassis centre. Height is 18 cm above base_footprint; yaw is 180 degrees in software.

Checkpoint	Pass condition
Position	x=-0.05 m from chassis centre
Height	z=0.18 m
View	Nothing crosses the 360-degree scan plane

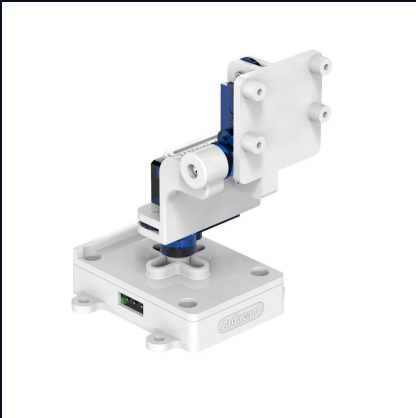


Mechanical assembly - camera and environment

Step 6 - Assemble camera pan/tilt

Build the Arducam bracket without overtightening servo shafts. Fit Camera Module 3 so the image is upright. Route the CSI ribbon through a loose service loop that permits full pan and tilt without pulling.

Checkpoint	Pass condition
Ribbon	Blue/contact orientation matches the Jetson carrier connector
Movement	Full commanded range without tension or collision
Default	Camera returns to saved centre/up viewing position after service start



Step 7 - Place environment sensors

Mount BME680 outside the enclosure in moving ambient air, protected from rain and direct exhaust/Jetson heat. Mount AMG8833 inside with an unobstructed thermal view. Do not swap their physical roles.

Checkpoint	Pass condition
BME680	Outside; bus 7; address 0x77
AMG8833	Inside; bus 7; address 0x69 or 0x68
Heat	Neither sensor touches Jetson heatsink or motor driver



Step 8 - Place radar, GNSS and antennas

Mount RD-03D facing the monitored direction with no metal immediately in front. Place the L76K antenna outside and sky-facing. Keep GNSS, Wi-Fi and cellular antennas separated from motor cables and the BMS.

Checkpoint	Pass condition
Radar	Front face unobstructed; UART cable labelled
GNSS	Antenna connector fully seated; module uses USB-to-TTL

Checkpoint	Pass condition
Antennas	Outside enclosure and strain relieved



Complete verified wiring map

Device	Power	Data connection	Verified software identity
Jetson Orin Nano Super	Dedicated regulated input matching kit	Main computer	jetson@192.168.1.14 / Tailscale when active
Yahboom motor board	Motor/logic supply per board label	USB CH340/CH341	/dev/yahboom
M1 motor/encoder	Yahboom M1	Controller encoder input	Front-right; + forward; 4048.7 counts/rev
M2 motor/encoder	Yahboom M2	Controller encoder input	Front-left; - forward; 3300.6 counts/rev
M3 motor/encoder	Yahboom M3	Controller encoder input	Back-right; - forward; 4080.1 counts/rev
M4 motor/encoder	Yahboom M4	Controller encoder input	Back-left; + forward; 2697.8 counts/rev
RPLIDAR A1	Via supplied USB adapter	USB CP2102	/dev/serial/by-id/...CP2102...; 115200
IMX708 camera	CSI interface	Jetson CSI ribbon	camera service / ROS image topic
Arducam pan/tilt	Separate 3.6-4.8 V servo rail; common GND	I2C bus 7, address 0x40	atlas-arducam-ptz.service
BNO08X IMU	3.3 V logic per breakout	Jetson I2C SDA/SCL	address 0x4B
BME680	Breakout-rated 3.3/5 V	I2C bus 7	address 0x77
AMG8833	Breakout-rated supply	I2C bus 7	address 0x69 or 0x68
RD-03D radar	Per module label; common GND	UART: radar TX->Jetson RX pin 10; radar RX->Jetson TX pin 8	/dev/ttyTHS1
L76K GNSS	From USB-to-TTL adapter	UART through USB-to-TTL	Do not share radar pins 8/10
Ultrasonic controller	Arduino supply	USB	Arduino UNO WiFi R4 by-id; 115200
ESP32-S3 speaker	USB-C or supported battery input	USB plus Wi-Fi	Voice companion service
CrowPanel	Its own supported supply/battery	Wi-Fi to ATLAS dashboard/API	Wireless controller; no direct motor wire

Jetson 40-pin header connections

Physical pin	Signal	ATLAS use	Connect
1	3.3 V	Logic reference only	Only breakout boards explicitly rated for 3.3 V
3	SDA	Shared I2C bus	BNO08X / Arducam control / BME680 / AMG8833 as configured
5	SCL	Shared I2C bus	Same I2C devices
6 or other GND	Ground	Common signal ground	Sensor grounds and regulated-rail common
8	UART TX	RD-03D receive	Jetson TX -> radar RX
10	UART RX	RD-03D transmit	Jetson RX <- radar TX

NO GPIO POWER

Do not connect motor, steering servo, pan/tilt servo, LiDAR motor, speaker amplifier or CrowPanel power to Jetson GPIO pins. Use the correct regulated rail and join grounds only where required.

I2C rules

Power off before changing I2C wiring. SDA connects to SDA and SCL to SCL. All devices share ground. Confirm every address with `i2cdetect` before enabling services. Address conflicts must be resolved with the installed mux or hardware address selection; never connect two active devices with the same fixed address on one bus.

USB rules

Use stable `by-id` or `udev` names, not assumed `ttyUSB` numbers. The motor board, LiDAR, ultrasonic Arduino, L76K USB-to-TTL and voice board must be separately labelled. Secure every USB plug against rover vibration.

Power distribution and emergency stop

Path	Required arrangement
Battery positive	Battery -> correctly rated fuse -> master disconnect/E-stop power contact -> distribution
Motor rail	Distribution -> Yahboom motor input using conductor/ferrule sized for stall current
Jetson rail	Distribution -> dedicated regulated Jetson input; never raw over-voltage
Servo rail	Dedicated regulated 3.6-4.8 V for Arducam digital servos; common signal ground
Logic/sensors	Regulated USB/3.3/5 V according to each board label
Ground	Common electrical reference without routing motor current through sensor ground wires
BMS	Installed exactly according to cell chemistry/count and DALY harness sequence
MEASURE FIRST	With the battery disconnected from electronics, verify pack voltage and polarity at the distribution input. ATLAS telemetry has shown approximately 12.2 V operating and 12.6 V full. If your measured pack or label is different, do not follow this power section blindly.

Power-off continuity checks

Meter test	Pass
Battery + to chassis	Open circuit / no unintended continuity
Battery - to common ground	Continuity only through intended ground path
Positive to negative at each unpowered rail	No near-zero-ohm short
E-stop open	Motor power path interrupted
Polarity	Every connector matches board + and - markings

First power-up - wheels lifted

Step 9 - Pre-power inspection

Lift all four wheels securely. Centre steering by hand only while unpowered. Disconnect motor outputs if any connector identity remains uncertain. Confirm the E-stop physically removes motor power.

Checkpoint	Pass condition
Battery	Correct voltage and polarity
Cooling	Jetson fan unobstructed
Wiring	No exposed conductor; no cable in wheel/linkage
Operator	Finger on E-stop; no person in wheel plane

Step 10 - Boot Jetson

Apply logic/Jetson power, then motor power according to the installed master switch sequence. Wait for Ubuntu and ATLAS services. Do not command movement during boot.

Checkpoint	Pass condition
Network	Dashboard or SSH reachable
Temperature	No rapid rise; fan operating
Services	Base, sensors, Nav2, SLAM, camera and dashboard active
Autonomy	Explore Lite remains inactive after boot

Useful check	Expected
Dashboard	Live CPU/GPU/RAM, battery and sensor freshness
RPLIDAR	Approximately 7-10 Hz scan; stable laser_frame
Xbox	/dev/input/js0 present
Motor command	Zero until LB/deadman plus stick input
Map	No autonomous goal without explicit command

Lifted-wheel motor acceptance

Test one motor at a time. Stop fully between tests. The expected forward direction is toward the physical front of the rover.

Order	Channel	Wheel	Forward sign	Acceptance
1	M1	Front-right	+	Wheel forward; encoder increases
2	M2	Front-left	-	Wheel forward; normalized encoder distance increases
3	M3	Back-right	-	Wheel forward; normalized encoder distance increases
4	M4	Back-left	+	Wheel forward; encoder increases

STOP ON ERROR

If any wheel turns backward, another wheel moves, encoder data stays at zero, or the controller becomes hot, stop immediately. Correct the labelled motor/encoder channel; do not hide wiring errors with random software sign changes.

Verified calibration baseline

Wheel	Counts/revolution	Physical location
M1	4048.7	Front-right
M2	3300.6	Front-left
M3	4080.1	Back-right
M4	2697.8	Back-left

After passing, test front and rear steering separately while wheels remain lifted. Confirm neutral alignment and unrestricted left/right movement. Then test camera pan and tilt without the CSI ribbon becoming tight.

Sensor acceptance before ground use

Sensor	Pass condition	If failed
LiDAR	Live /scan; no chassis self-return; TF x=-0.05, z=0.18, yaw=pi	Check USB, rotation, mount and self-filter
BNO08X	Fresh heading/orientation at 0x4B	Check I2C, address, mount orientation
Encoders	All four distances change with the correct wheel	Check motor/encoder connector and controller
Ultrasonic	Front/left/right values react to a flat target	Check Arduino USB, crossed channels and sensor face
RD-03D	Targets/detections update on UART	Check 5V/GND per module, crossed TX/RX, /dev/ttyTHS1
L76K	NMEA data and satellites through USB-to-TTL	Move antenna outdoors; check TX/RX and 9600 default
BME680	Outside temp/humidity/pressure/gas update at 0x77	Check bus 7 and placement
AMG8833	64 thermal pixels update at 0x69/0x68	Check bus 7 and address
Camera	Live image; pan/tilt; AI overlay as selected	Check CSI orientation, service and I2C controller
BMS	Voltage/current/SOC and charge/discharge state	Check BMS power/Bluetooth telemetry path
GROUND GATE	Do not place ATLAS on the ground until motors, steering, encoders, E-stop and obstacle sensors all pass with the wheels lifted.	

Ground commissioning

Test	Setup	Pass
Manual light mode	Clear floor; 10% mode; operator at E-stop	Smooth forward, reverse and steering; immediate release stop
Boost mode	Only after light mode passes	Approximately 30% operator mode; no wheel rubbing or brownout
20 cm odometry	Mark start/end on floor	Physical movement and odometry agree within commissioning tolerance
Obstacle stop	Flat target; low speed	LiDAR primary safety stops before contact; ultrasonic is secondary
Return home	Set home, move a short verified distance	Returns physically near saved origin; keep route clear
Mapping	Initial 30 cm clearance; clear start	START_POSE VALID; one stable frontier goal; map grows
Map save	Stop exploration normally	atlas_latest.yaml and .pgm receive fresh timestamps
Reboot	Park and reboot	Required services return; Explore remains inactive; no unexpected motion

ATLAS passed the primary short-move, return-home, bounded mapping, collision abort, automatic safety stop, map-save and reboot/autostart sequence on 06 August 2026. Longer full-room endurance mapping remains a supervised field-refinement task.

AUTONOMY	Use a bounded safety-stop timer during early mapping. If Wi-Fi/Tailscale is lost during autonomy, use the physical E-stop. Never depend on a remote web button as the only stop mechanism.
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Normal operation - quick start

Action	Procedure
Power on	Inspect rover -> release E-stop -> power Jetson -> wait for dashboard healthy state
Manual drive	Xbox connected -> hold LB deadman -> left stick drive/steer; RB enables boost
Camera	Use remote/dashboard pan-tilt; enable AI/face tracking only when needed
Foxglove	Connect to bridge on port 8765; view map, scans, camera and diagnostics
Web dashboard	Open ATLAS address on the local network/Tailscale; use manual control with live safety state
Start mapping	Confirm clear start -> START_POSE VALID -> start exploration -> supervise
Stop mapping	Stop exploration -> wait for map saver -> verify atlas_latest files
Park	Return home or manual park -> stop mux/exploration -> verify zero command
Shutdown	Save map -> stop motion -> normal sudo/system shutdown -> remove power after halt

Drive modes

Mode	Purpose	Typical use
Light 10%	Fine/indoor positioning	First test and tight spaces
Boost 30%	Higher manual drive authority	Open, clear floor only
Nav2	Autonomous planned velocity	Supervised goals and mapping
Stopped	Mux inactive or watchdog zero	Parking, wiring and fault response

Troubleshooting - fastest safe checks

Symptom	First check	Safe correction
Remote does not move rover	Xbox /dev/input/js0; LB held; mux active	Restart remote and mux; keep Explore off
Only one wheel moves	Motor labels, connector seating, encoder telemetry	Lift wheels; test M1-M4 individually
Rover twitches but cannot move	Battery voltage, motor rail, mechanical drag, PWM floor	Fix power/mechanics; do not exceed safe calibrated PWM
Camera moves wrong direction	Bracket orientation and saved servo limits	Recalibrate limits; never force servo
No camera feed	CSI ribbon orientation and camera service	Power off before reseating CSI
Map not visible	SLAM node, /scan and TF timestamps	Stabilize TF; restart SLAM then Nav2
START_POSE BLOCKED	Nearby obstacle/self-return/stale map	Keep autonomy off; reposition and recreate fresh map
Repeated frontier goals	Explore Lite build/version	Deploy patched goal-hold Explore Lite
GPS sees satellites but no fix	Antenna sky view, NMEA fix fields	Wait outdoors; inspect antenna and USB-TTL
High CPU	SLAM/Nav2 plus camera/Foxglove/dashboard	Keep safety sensors active; reduce visualization load/priority
Network lost	Wi-Fi route/Tailscale	Physical E-stop during autonomy; reconnect locally
Battery data absent	BMS awake/telemetry path	Stop high-load tests until voltage/current is known

Disassembly and reassembly

Disassemble in the reverse build order. Photograph and label every connector before removal. Keep calibration-sensitive assemblies together whenever possible.

Before removing	Required action
Battery/BMS	Shut down Jetson; disconnect charger; open master disconnect; verify zero voltage at load
CSI camera	Power fully off; release connector latch; pull ribbon straight
I2C sensors	Record address, bus and connector orientation
Motors/encoders	Label M1 FR, M2 FL, M3 BR, M4 BL at both ends
LiDAR	Measure/mark x=-0.05 m and z=0.18 m reference before removal
Steering	Mark neutral horn/link positions
Antennas	Release connector vertically using proper tool; never pull coax

Reassembly acceptance

Any change to motor, wheel, shaft, steering, LiDAR or IMU mounting invalidates the corresponding calibration. Repeat power-off checks, lifted-wheel direction/encoder tests, TF verification, measured ground move and clear-start footprint check before autonomous use.

DO NOT SKIP

A rover that powers on is not automatically safe. Reassembly is complete only after the full sensor and motion acceptance gates pass.

Verified software/service baseline

Function	Service / interface
Motor telemetry	rover-base-telemetry.service
LIDAR	atlas-lidar.service + atlas-scan-filter.service + atlas-laser-tf.service
IMU/EKF	atlas-imu.service + atlas-ekf.service
Ultrasonic	atlas-ultrasonic.service
SLAM/Nav2	atlas-slam-fast.service + atlas-nav2.service
Manual priority	atlas-remote.service + atlas-cmd-vel-mux.service
Mapping	atlas-explore.service (explicit start only)
Camera/AI	atlas-camera.service + atlas-ai-annotator.service
Mission controls	atlas-mission-control.service
Foxglove	atlas-foxglove.service, port 8765
Web dashboard	rover-status-web.service
Map files	/home/jetson/project_atlas/maps/atlas_latest.yaml/.pgm

Commissioned Nav2 geometry

Parameter	Value
Footprint	[[0.25,0.18],[0.25,-0.18],[-0.25,-0.18],[-0.25,0.18]]
Inflation radius	0.28 m
Cost scaling factor	15.0
Obstacle min range	0.05 m
Behavior tree	atlas_ackermann_fail_stop.xml
Explore behavior	Hold one active frontier goal; guarded recovery after true aborts

Product photo and technical sources

Photos are included for identification and are copyright of their respective manufacturers/vendors. Always consult the current datasheet supplied with the exact purchased revision.

Component	Reference URL
Jetson deployment kit	thinkrobotics.com/products/nvidia-jetson-orin-super-nano-deployment-kit-made-in-india
Chassis	amrobotics.in/products/rover-4wdxl60r-3-in-1-do-it-yourself-chassis-kit
125 mm wheels	amrobotics.in/products/125mm-heavy-duty-high-density-rc-rubber-wheel-with-flange-for-6mm-shaft-motors
RPLIDAR A1	slamtec.com/en/lidar/a1spec
Yahboom controller	category.yahboom.net/collections/accessory/products/ros-driver-board
Camera Module 3	raspberrypi.com/products/camera-module-3/
Arducam pan/tilt	rees52.com/products/arducam-upgraded-camera-pan-tilt-platform-compatible-with-raspberry-pi-nvidia-jetson-na no-xavier-nx-rs2854
RD-03D	docs.ai-thinker.com/en/Rd-03D_V1/
L76K	waveshare.com/product/l76k-gps-module.htm
BME680	adafruit.com/product/3660
AMG8833	learn.adafruit.com/adafruit-amg8833-8x8-thermal-camera-sensor
ESP32-S3 speaker	thinkrobotics.com/products/esp32-s3-ai-smart-speaker-development-board-onboard-dual-microphone-array
CrowPanel	elecrow.com/crowpanel-advanced-10-1inch-esp32-p4-hmi-ai-display-1024x600-ips-touch-screen-wifi-6.html
DALY BMS	dalybms.com/product/

END OF MANUAL